SimpleGrid problems compiled to d-DNNF with d4 (projected, no evidence)

Instance	Horizon	Circuit file	d4 time (s)	Total time (s)	CNF variables	CNF clauses	Projected variables	Circuit nodes	Circuit edges	Models (checked)	Tries
grid3x3-1	5	grid3x3-1.dnnf	0.03	1.9	141	818	98	43	81	18	1
grid3x3-2	5	grid3x3-2.dnnf	0.03	1.9	141	818	98	45	85	18	1
grid3x3-3	5	grid3x3-3.dnnf	0.04	1.9	141	818	98	45	85	18	1
grid3x3-4	5	grid3x3-4.dnnf	0.04	1.9	141	818	98	45	85	18	1
grid4x4-1	7	grid4x4-1.dnnf	0.04	1.9	400	2,609	291	228	451	140	1
grid4x4-2	7	grid4x4-2.dnnf	0.04	1.9	400	2,609	291	228	451	140	1
grid4x4-3	7	grid4x4-3.dnnf	0.04	1.9	400	2,609	291	228	451	140	1
grid4x4-4	7	grid4x4-4.dnnf	0.04	1.9	400	2,609	291	228	451	140	1
grid5x5-1	9	grid5x5-1.dnnf	0.14	2.3	865	5,994	644	2,606	5,208	2,100	1
grid5x5-2	8	grid5x5-2.dnnf	0.06	1.9	760	5,248	564	393	781	315	1
grid5x5-3	9	grid5x5-3.dnnf	0.10	2.1	865	5,994	644	1,032	2,059	1,372	1
grid5x5-4	8	grid5x5-4.dnnf	0.05	1.7	760	5,248	564	399	793	280	1
grid6x6-1	11	grid6x6-1.dnnf	1.54	7.5	1,596	11,477	1,205	24,754	49,539	36,510	1
grid6x6-2	11	grid6x6-2.dnnf	2.02	9.6	1,596	11,477	1,205	29,556	59,122	28,302	1
grid6x6-3	10	grid6x6-3.dnnf	0.28	2.9	1,440	10,333	1,085	3,860	7,717	3,570	1
grid6x6-4	11	grid6x6-4.dnnf	1.63	8.5	1,596	11,477	1,205	26,389	52,808	28,302	1
grid7x7-1	12	grid7x7-1.dnnf	6.27	29.5	2,436	17,936	1,854	73,184	146,525	96,558	1
grid7x7-2	12	grid7x7-2.dnnf	4.54	20.9	2,436	17,936	1,854	57,227	114,500	65,472	1
grid7x7-3	12	grid7x7-3.dnnf	6.60	27.9	2,436	17,936	1,854	91,315	182,661	86,856	1
grid7x7-4	12	grid7x7-4.dnnf	8.67	40.2	2,436	17,936	1,854	109,265	218,648	103,620	1
grid8x8-1	14	grid8x8-1.dnnf	41.09	284.6	3,808	28,561	2,919	340,768	682,049	1,170,962	1
grid8x8-2	15	grid8x8-2.dnnf	281.95	1173.0	4,096	30,753	3,143	1,044,687	2,093,264	7,856,368	8
grid8x8-3	15	grid8x8-3.dnnf	202.60	563.3	4,096	30,753	3,143	794,002	1,589,508	4,160,312	10
grid8x8-4	15	grid8x8-4.dnnf	121.20	553.1	4,096	30,753	3,143	963,185	1,929,510	8,536,710	6
grid9x9-1	16	none	error	–	5,616	42,712	4,328	d4 crashed after 0.36 s	–	–	6
grid9x9-2	16	none	error	–	5,616	42,712	4,328	d4 crashed after 0.33 s	–	–	6
grid9x9-3	16	none	error	–	5,616	42,712	4,328	d4 crashed after 38.08 s	–	–	6
grid9x9-4	16	none	714.78	timeout (30 min)	5,616	42,712	4,328	2,192,470	4,390,807	–	8

Each problem's evidence-free recognition theory — the problem at its recognition horizon, with the goal literals as the hypotheses, as recognize.sh builds it — compiled by marginals.sh `--solver` d4 `--project`. The projection keeps the drive actions and the final-slice goal atoms, and FiFO first verifies that they determine every other atom, so the projected circuit gives exact counts; any pinned observations are unit literals over the kept actions, so one saved circuit answers every evidence file by conditioning, without recompiling. CNF variables and clauses are those of the ground theory d4 is given; projected variables are the ones it keeps. d4 time is the d4 run alone; total time is the whole marginals.sh run (reading the theory, the projection check, d4, importing and saving the circuit, and the marginals of the unconditioned circuit). Circuit nodes and edges are those of d4's decision-DNNF, before FiFO smooths it. Times are wall-clock, 1 or 2 compilations at a time (grid8x8-2, grid8x8-3, grid8x8-4, grid9x9-1, grid9x9-2, grid9x9-3, grid9x9-4 one at a time: two 8 GB SBCL imports at once swap on this 16 GB machine), each limited to 30 minutes. SBCL ran with an 8 GB heap (grid9x9-4: 12 GB: with 8 GB, SBCL's garbage collector failed while importing its circuit) and a 1 GB control stack, d4 with a 64 MB stack (the defaults, 1 GB, 2 MB and 8 MB, are too small from 7×7 or 9×9 on). The circuits were imported by FiFO with its d4 importer rewritten to use bit-vector scopes and iterative traversal (branch ddnnf-import-bitvector-scopes); the released importer runs out of memory on circuits of about a million nodes. Every circuit kept was checked: its model count (all literals set to 1 in the smoothed circuit) equals the number of truck walks the theory allows, counted independently by dynamic programming over the grid (H-1 steps, each a drive to a neighbouring cell or a wait, from p1-1 to a goal cell), a count that agrees with sharpSAT on every problem up to 7×7 (sharpSAT did not finish an 8×8 problem in hours). The check is needed: from 8×8 on, d4 sometimes crashes (bus error or segmentation fault) and sometimes returns a wrong circuit without any error, and repeated runs on the same CNF give different circuits, so each problem was compiled until a run passed. Tries counts the runs of d4 on a problem, failed ones included, and the times are those of the run that passed. For grid9x9-4, FiFO's circuit-size cap (*ddnnf-node-limit*, 2,000,000 nodes) was raised to 20,000,000: the smoothed import of its d4 circuit exceeds the default. In the run reported, d4 was given extra options: "`--cache-store-strategy` all `--cache-size-first-page` 268435456 `--cache-size-additional-page` 268435456" for grid8x8-2, grid9x9-1, grid9x9-2, grid9x9-3, grid9x9-4; "`--cache-store-strategy` all" for grid8x8-3 (every other problem: d4's defaults). Storing every clause in a cache entry, and cache pages smaller than the default 4 GB, made d4 miscompile less often, though not never. "error": no run finished — d4 crashed every time. A timeout with circuit sizes: d4 finished, but importing and saving its circuit did not within the limit. No 9×9 problem compiled within the limit, so larger grids were not tried.